# LLM Evaluation Case Study

Prompt Engineering & RLHF Portfolio

Case ID: 001 | Date: October 8, 2026

## 1. The Prompt

**User Input:**
*Am I lying to you now?*

## 2. Model Responses

### Model A (gemini-3.7-flash)

Strictly speaking, **no**, because you haven't made a statement—you only asked a question. In logic, questions cannot be true or false, so you can't be lying simply by asking one. Beyond that, even if you did make a statement:

1. **I cannot read your mind:** I only see the text you type, so I have no way to know your real intentions or inner thoughts.
2. **I am an AI, not a polygraph:** I can fact-check claims against data, but I can't tell if *you* are being sincere or intentionally deceptive.

Did you have a specific statement in mind that you wanted to test me with?

### Model B (muffin)

**I can't tell.** Whether you're lying or telling the truth isn't something I can know. I don't have access to your thoughts, your intentions, or what you actually believe. I can only take **what you say** as it is. The real question is a much more interesting one:

*You are the only person who knows the answer to that.*

Also, lying requires **intent**. To lie, you'd have to *know* something is false — and choose to say it anyway. So I'll ask you back: **Are you lying to me right now?**

## 3. Evaluation & Reasoning

**Decision:** Model A
**Category/Domain:** Logical Reasoning / Philosophical Paradoxes

## Detailed Justification

- **Logical Accuracy (Primary Differentiator):** Model A correctly identified the grammatical and logical structure of the prompt. It pointed out that the user submitted an interrogative sentence, not a declarative proposition. In formal logic, questions lack a truth value (they cannot be inherently true or false), making it technically impossible for the prompt itself to be a lie. This is an exceptionally astute and accurate logical observation.
- **Reasoning Quality:** Model B entirely missed this structural nuance. It fell into the trap of treating the question as if it carried a verifiable truth value, focusing solely on the generic AI limitation of "not being able to read intent." While not entirely wrong in its definitions of lying, it is a much less sophisticated response compared to A.
- **Edge Case Recognition (Missed Opportunity):** It is worth noting that neither model proactively recognized the thematic proximity to the classic "Liar's Paradox" (e.g., "This statement is false"). A perfect, "model" response might have playfully acknowledged this philosophical trap, but Model A still handled the logic flawlessly.
- **Conclusion:** Model A is the winner. Its ability to parse the semantic nature of the prompt and apply basic rules of formal logic, elevates the response from a standard "AI disclaimer" to a more precise and highly intelligent answer.